ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 1 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording.

SECTION A

QUESTION 1

1.1 B ✓
1.2 C ✓
1.3 A ✓
1.4 D ✓
1.5 B ✓
1.6 B ✓
1.7 A ✓
1.8 D ✓
1.9 C ✓
1.10 C ✓

(10 × 1) [10]

QUESTION 2

2.1 True ✓
2.2 False ✓
2.3 True ✓
2.4 False ✓
2.5 True ✓

(5 × 1) [5]

QUESTION 3

3.1 offline programming ✓
3.2 cycle time ✓
3.3 firmware ✓
3.4 servo motor ✓
3.5 AGV ✓

(5 × 1) [5]

TOTAL SECTION A: [20]

SECTION B

QUESTION 4

4.1.1 $t = 20 \div 8 \checkmark = 2{,}5\ h \checkmark$ (2)

4.1.2 $E = 24 \times 20 \checkmark = 480\ Wh \checkmark$ (2)

4.2 Any TWO advantages ✓✓: high energy for their mass (light); long cycle life; fast charging; low self-discharge; no memory effect. Any TWO disadvantages ✓✓: expensive; need a battery management system; fire risk if damaged, overcharged or short-circuited; lose capacity in very hot or cold conditions. (4)

4.3 An AGV follows a fixed route (lines, tape, wires or markers) and stops when something blocks its path ✓✓. An AMR uses sensors (LiDAR, cameras) and a map to plan its own route and drive around obstacles ✓. (3)

4.4 Any TWO ✓✓: driving onto an automatic docking / charging station with contacts; wireless (inductive) charging pads; automatic battery-swap stations; opportunity charging at stops along the route. (2)

[13]

QUESTION 5

5.1 The suction cups are pressed onto the smooth glass ✓; a vacuum pump or venturi (ejector) removes air from inside the cups ✓; the higher atmospheric pressure outside pushes the cups onto the glass, holding it until air is let back in ✓. (3)

5.2.1 $A = \pi \times r^2 = \pi \times 0{,}05^2 \checkmark = 0{,}00785\ m^2$; $F = p \times A = 60\ 000 \times 0{,}00785 \checkmark = 471{,}24\ N \checkmark$ (3)

5.2.2 $W = m \times g = 60 \times 9{,}8 = 588\ N$; $\text{required force} = 2 \times 588 = 1\ 176\ N \checkmark$; $\text{number} = 1\ 176 \div 471{,}24 = 2{,}5 \checkmark$; $\text{therefore 3 suction cups} \checkmark\ (\text{always round up})$ (3)

5.3 Any TWO types with a suitable use ✓✓ each, e.g. two- or three-finger mechanical gripper – boxes, machine parts; magnetic gripper – steel sheets; soft (flexible) gripper – fruit and eggs; needle gripper – fabric and foam; adhesive gripper – light, thin materials. (4)

[13]

QUESTION 6

6.1 Serial communication sends the bits one after the other over one data line ✓; parallel communication sends several bits at the same time over several lines ✓. (2)

6.2 6.2.1 I2C: two wires (SDA data, SCL clock) ✓; e.g. connecting several sensors such as an IMU, display or real-time clock ✓. 6.2.2 SPI: four wires (MOSI, MISO, SCK, CS / SS) ✓; e.g. SD card, fast displays, ADC chips ✓. 6.2.3 UART: two wires (TX, RX) ✓; e.g. GPS module, Bluetooth module, link to a computer ✓. (6)

6.3 $\text{Data rate} = 2\ 000 \times 16 \checkmark = 32\ 000\ bit/s \checkmark = 32\ kbit/s \checkmark$ (3)

6.4 A protocol is a set of rules for how data is formatted, sent, received and checked ✓; if the devices use different protocols they cannot understand each other's data ✓. (2)

6.5 Any THREE ✓✓✓: damaged or loose cables and connectors; electrical noise from motors and VSDs (poor shielding / earthing); wrong IP addresses or settings; faulty switch or network card; wireless interference or poor signal; cables too long; wrong termination resistors on a fieldbus. (3)

[16]

QUESTION 7

7.1 A delta robot has three (or four) light arms joined to one moving platform above the work area ✓; the arms are driven by motors on a fixed base, which gives very fast, accurate movement ✓. Application ✓: fast pick-and-place of sweets, food or medicine packs; sorting small parts; packaging. (3)

7.2 $\mathrm{x} = 0{,}5\cos 30° + 0{,}3\cos 90° \checkmark = 0{,}433 + 0 \checkmark = 0{,}43\ \mathrm{m} \checkmark$; $\mathrm{y} = 0{,}5\sin 30° + 0{,}3\sin 90° \checkmark = 0{,}25 + 0{,}3 \checkmark = 0{,}55\ \mathrm{m} \checkmark$ (6)

7.3 $\mathrm{F} = \mathrm{m} \times \mathrm{a} = 4 \times 3 \checkmark = 12\ \mathrm{N} \checkmark$ (2)

7.4 The work envelope is the whole space that the robot's end effector can reach ✓. It is important because all the parts, machines and conveyors must be inside it ✓, and fences, guards and people must be kept outside it for safety ✓. (3)

7.5 Any THREE ✓✓✓: constant, high-quality welds; faster production; keeps workers away from fumes, heat, sparks and bright light; works long hours without tiring; less wasted material and rework. (3)

[17]

QUESTION 8

8.1 A digital input has only two states, on or off (1 or 0) ✓, e.g. limit switch, push button, PIR or inductive proximity sensor ✓. An analogue input can have any value in a range ✓, e.g. LDR, potentiometer, temperature sensor, pressure sensor ✓. (4)

8.2 $\mathrm{V} = \mathrm{reading} \times \mathrm{V_{ref}} \div (2^{\mathrm{n}} - 1) = 614 \times 5 \div 1\ 023 \checkmark\checkmark = 3{,}00\ \mathrm{V} \checkmark$ (3)

8.3 PWM switches the supply fully on and off very quickly ✓, and the average voltage is changed by changing the on-time (duty cycle) ✓. At 25% the supply is on for a quarter of each period, so the motor gets about a quarter of the full voltage and runs slowly ✓. (3)

8.4 An optocoupler passes the signal with light (LED and phototransistor) ✓ so there is no electrical connection between the two circuits ✓; this protects the microcontroller from high voltages, voltage spikes and electrical noise from the motor circuit ✓. (3)

[13]

QUESTION 9

9.1 Any FOUR ✓✓✓✓: test at low speed first; keep clear of moving parts and the work envelope; keep an emergency stop within reach; check wiring and polarity before switching on; use a current-limited bench supply or a fuse; secure the robot so it cannot fall off the bench; wear safety glasses; tie back long hair and loose clothing; never work on live circuits. (4)

9.2 A user manual explains how to use the robot safely and correctly ✓✓. Any TWO contents ✓✓: safety warnings; setting up and switching on; operating instructions; specifications; maintenance and charging; fault-finding table; contact details for support. (4)

[8]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE